

Feedback on C142 - Discrete Mathematics - 2021

This year's exam has been of average difficulty with respect to previous years, but had certain parts that a lot of students found really difficult.

The main point I would like to make is that many students showed poor time management. Very few students answered all the parts, with too many answering only two parts for Q2.

Many spent far too much time on part 1(d), time that had better be invested in the other parts. Please bear in mind that we cannot give any marks to an empty answer, and that spending a lot of time on one part cannot compensate for not answering another.

Question 1 Average performance: A.

a) This part asked for some set construction, testing the understanding of the basic notions; most students got this right.

b) This was a variant of an exercise from the Xmas test; most students got this right.

c) This asked to prove a subset relation between two sets, and required understanding of what steps are allowed in proofs. Most students did reasonably well, but skipped steps in the proof.

d) This was a tricky part, and most students over invested in writing all details of the answer, not seeing that part of the answer could be found in the notes.

e) This part asked to show that a relation was a partial order; on average, the students knew what to do and how to do it.

Question 2 Average performance: D.

a) This part was done well; some students forgot to properly motivate their answer, although many students did not correctly check the formal definitions involved.

b) This was a variant of coursework and was done really well; most students got this right.

c) This part asked to understand a formal definition of a relation, and study its properties; about half of the students got this right.

d) I was expecting more for this part, seen that it appeared almost identically in the notes; most students misread the question, thereby losing marks.

e) This part was intended as the 'difficult one', and it has proven to be that. Many students not even attempted this part; some that did, got confused when trying to build a diagonalisation argument.

The result follows from a result in the notes, but very few students realised that.